COMPUTE ALLOCATION REPORT

CloudLab & Chameleon Cloud

Free bare-metal NSF testbeds for systems research and experimental GPU workloads

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Program Overview

CloudLab and Chameleon Cloud are the two flagship National Science Foundation (NSF) experimental research testbeds for cloud, systems, and networking research in the United States. Both provide *bare-metal* access to whole servers (no hypervisor), root privileges, full network reconfiguration, and custom OS images, and both are free to U.S.-affiliated researchers who agree to publish their results in the open. They are not production HPC clusters. They are instruments for conducting reproducible computer-science experiments on real hardware, and their allocation, scheduling, and storage models reflect that experimental orientation.

CloudLab is operated by the University of Utah in partnership with the University of Wisconsin-Madison, Clemson University, and the University of Massachusetts Amherst, with additional federated sites at APT (Utah) and on the Mass Open Cloud. The platform exposes roughly 1,800 servers across the four primary sites, with continuously refreshed hardware generations and an emphasis on cloud-stack research, software-defined networking, distributed systems, kernel and OS work, and increasingly ML-systems experimentation. Users instantiate *profiles* (declarative experiment specifications stored on GitHub) that allocate one or more bare-metal nodes, optionally wire them into custom topologies through the underlying Emulab fabric, and boot arbitrary disk images.

Chameleon Cloud is led by the Texas Advanced Computing Center (TACC) and the University of Chicago, with additional sites at Northwestern University (StarLight) and the National Center for Atmospheric Research (NCAR). Like CloudLab, Chameleon offers bare-metal reservations on heterogeneous generations of hardware. Unlike CloudLab, Chameleon places strong emphasis on *reproducibility tooling*. The Trovi catalog hosts shareable Jupyter artifacts that bundle a reservation script, an image, and an analysis notebook, so a paper's reviewer can re-run the entire experiment with a single click against a fresh lease. Chameleon also operates a Swift-based object store for persistent data, an OpenStack KVM partition (CHI@Edge, CHI@KVM) for lightweight workloads, and since 2023 has expanded an explicit *AI research* track that has grown into the “Chameleon 2.0” AI hardware program.

Both projects are funded under NSF's CISE Community Research Infrastructure program. CloudLab's current renewal runs through 2026; Chameleon's third phase (“Chameleon 2.0”) was funded in 2024 and runs through 2028 with explicit AI-systems hardware investments.

CloudLab Hardware

CloudLab's hardware is highly heterogeneous and rotates frequently. The table below lists representative GPU-bearing node types verified against cloudlab.us/hardware. Many non-GPU node families (c220g1, c220g2, m400, xl170, etc.) exist alongside these and are useful for distributed-systems and networking experiments.

Cluster	Node	GPUs	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	CPU	RAM (GB)	NIC
Utah	d740	1× A100 PCIe	40	312	1.55	2× Xeon Gold 6336Y (24c)	128	25 GbE
Utah	d430	—	—	—	—	2× Xeon E5-2630v3 (8c)	64	10 GbE
Wisconsin	c6525-100g	—	—	—	—	2× AMD EPYC 7402P (24c)	128	2× 100 GbE
Wisconsin	c6525-25g	—	—	—	—	2× AMD EPYC 7402P (24c)	128	2× 25 GbE
Wisconsin	c220g5	optional 1× V100	16/32	125	0.90	2× Xeon Silver 4114 (10c)	192	10 GbE
Wisconsin	c4130	4× V100 SXM2	16	125	0.90	2× Xeon E5-2680v4 (14c)	256	10 GbE
Wisconsin	c240g5 / T4	1× T4	16	65	0.32	2× Xeon Silver 4114 (10c)	192	10 GbE
Clemson	c8220	—	—	—	—	2× Xeon E5-2660v2 (10c)	256	10 GbE

Cluster	Node	GPUs	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	CPU	RAM (GB)	NIC
Clemson	r320	—	—	—	—	1× Xeon E5-2450 (8c)	16	1 GbE
Clemson	ibm8335 (P9)	4× V100 SXM2	16	125	0.90	2× POWER9 (16c)	512	25 GbE
Mass	rs440xl	optional 1× P100	16	21	0.73	2× Xeon Gold 6240R (24c)	384	25 GbE
Mass	sm220u	—	—	—	—	2× Xeon Silver 4214R (12c)	192	25 GbE

GPU inventory on CloudLab is genuinely scarce by AI-cluster standards. The A100 d740 nodes at Utah and the V100 c4130/ibm8335 nodes at Wisconsin and Clemson are the high-end tier, and they routinely show full reservation calendars 1–2 weeks ahead. The Mass site is mostly CPU and storage oriented. CloudLab's strength is *network-attached experiments*, not raw GPU throughput.

Chameleon Hardware

Chameleon's hardware is documented at chameleoncloud.org/hardware, with each node grouped under a *node_type* string used in `blazar` lease requests. Representative GPU node types verified against the public hardware catalog appear below.

Site	Node type	GPUs	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	CPU	RAM (GB)	NIC
CHI@TACC	compute_gpu_p100	2× P100 PCIe	16	21	0.73	2× Xeon E5-2670v3 (12c)	192	10 GbE
CHI@TACC	compute_gpu_rtx_6000	2× RTX 6000	24	130	0.67	2× Xeon Gold 6240R (24c)	192	25 GbE

Site	Node type	GPUs	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	CPU	RAM (GB)	NIC
CHI@TACC	compute_liquid (mixed)	up to 8× A100/V100 (composable)	40/32	312/125	1.55/0.90	2× Xeon Gold 6230 (20c)	384	100 GbE
CHI@UC	compute_gpu_a100_nvlink	4× A100 SXM4 80GB	80	312	2.04	2× AMD EPYC 7763 (64c)	1024	2× 100 GbE
CHI@UC	compute_gpu_a100_pcie	2× A100 PCIe 40GB	40	312	1.55	2× AMD EPYC 7543 (32c)	512	100 GbE
CHI@UC	compute_gpu_v100	2× V100 PCIe	32	125	0.90	2× Xeon Gold 6126 (12c)	192	25 GbE
CHI@UC	compute_gpu_mi100	1× AMD MI100	32	185	1.23	2× AMD EPYC 7543 (32c)	256	25 GbE
CHI@NW	compute_gpu_p100_nvlink (StarLight)	4× P100 SXM2	16	21	0.73	2× Xeon E5-2680v4 (14c)	256	40 GbE
CHI@NCAR	compute_cascadelake_r	—	—	—	—	2× Xeon Gold 6248R (24c)	192	25 GbE

Chameleon 2.0 AI Hardware

Chameleon 2.0, the 2024–2028 phase, is rolling out an explicit AI hardware tier targeted at machine-learning systems research. Verified through the 2024–2026 hardware announcements, the planned and partially-deployed inventory includes the following.

Site	Node type / project	GPUs	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	Status
CHI@TACC	compute_gpu_h100 (Liqid composable)	8× H100 PCIe	80	756	2.00	Deployed 2024
CHI@UC	compute_gpu_h100_nvlink	4× H100 SXM5	80	989	3.35	Deployed 2025
CHI@UC	compute_gpu_grace	1× GH200 Grace-Hopper	96	989	4.00	Deployed 2025
CHI@UC	compute_gpu_mi210	2× AMD MI210	64	181	1.64	Deployed 2024
CHI@TACC	compute_gpu_h200 (announced)	4× H200 SXM5	141	989	4.80	Pilot 2026
CHI@NCAR	atmospheric AI testbed (NSF NCAR)	mixed A100/H100	40–80	312–989	1.55–3.35	Phased 2025–2026

The Liqid composable fabric at CHI@TACC is unique among NSF testbeds. It allows researchers to dynamically attach 1–8 GPUs to a single host over PCIe, which makes it ideal for studying GPU-disaggregation, heterogeneous scheduling, and oversubscription policies that cannot be expressed on a fixed-topology system.

Allocation Model and H200-Hour Equivalence

Both testbeds use a *reservation* model rather than a queue. A user requests a specific node type for a specific window; if the calendar is open the node is held exclusively for that user, booted into the chosen image, and released back to the pool when the lease expires.

CloudLab leases default to 16 hours and may be extended through the web UI up to roughly 7 days, with longer holds requiring a short justification emailed to the operations team. There is no explicit credit currency. Fairness is enforced by per-user concurrent-experiment caps and by the operations team intervening if a small group monopolizes scarce nodes.

Chameleon meters usage in *Service Units* (SUs), where 1 SU equals one node-hour regardless of node type. New projects receive an initial allocation, commonly 20,000 SUs for a six-month renewable term, with top-ups by request. Leases default to 1 day and may be extended up to 7 days at a time, with a hard ceiling near 30 days for advance reservation.

The H200-hour equivalence below converts a typical 20,000 SU project-year on Chameleon to dense FP16 H200-hours, summed over the GPUs attached to the reserved node and using $\text{hours} \times \text{GPUs} \times \frac{\text{TFLOPS}}{989}$ as the conversion. The "utilization caveat" matters more here than on a queued HPC system, because a researcher pays SUs for the entire wall-clock lease even when the GPUs sit idle during instrumentation, debugging, or kernel recompiles.

Reservation profile	Node-hours	GPUs / node	Per-GPU TFLOPS	H200-hr equivalent
Chameleon 20k SUs on a100_nvlink (4× A100 80GB)	20,000	4	312	25,228
Chameleon 20k SUs on h100_nvlink (4× H100 SXM5)	20,000	4	989	80,000
Chameleon 20k SUs on gh200 (1× GH200)	20,000	1	989	20,000
Chameleon 20k SUs on rtx_6000 (2× RTX 6000)	20,000	2	130	5,258
Chameleon 20k SUs on p100 (2× P100)	20,000	2	21	849
CloudLab d740 A100 (1 node, 1 wk × 26 wk)	4,368	1	312	1,378
CloudLab c4130 V100 (1 node, 1 wk × 26 wk)	4,368	4	125	2,208

A 20,000 SU pool on the H100 NVLink node corresponds to roughly 80,000 dense FP16 H200-hours of compute, which is competitive with mid-tier ACCESS or NAIRR awards. In practice, a Chameleon project rarely converts that favourably because the SU meter does not pause when nodes are idle.

Approval Timeline

Account creation on either testbed takes one to three business days, gated on institutional-email verification. Project creation is a separate step requiring a short scientific description, an open-science commitment, and (for student requesters) a faculty sponsor; project review usually completes within one week. Once a project is approved, reservations are made directly through the web UI or the API, with availability the only constraint. Popular GPU node types on both testbeds, particularly the A100 NVLink and H100 nodes on Chameleon and the d740 and c4130 nodes on CloudLab, can require booking 1–2 weeks ahead, especially around paper deadlines.

Eligibility

Both testbeds require a U.S.-based or U.S.-collaborating affiliation and an open-science research project. CloudLab's stated scope is *systems, networking, OS, and cloud-architecture experimental research*, and pure machine-learning training without an explicit systems contribution is technically out-of-scope, although enforcement is light when a project description frames the work in systems terms (scheduling, parallelism, memory hierarchy, communication, fault tolerance). Chameleon has been more permissive of AI workloads since 2023 and now actively recruits ML-systems projects under the Chameleon 2.0 AI track. International collaborators may be added to either project as users but cannot serve as the lead PI.

Things to Know

Researchers receive *full root* on every allocated node and are expected to treat each lease as a clean install. There is no shared cluster filesystem in the HPC sense. Home directories are small (typically a few GB on CloudLab, similarly modest on Chameleon login nodes), and any large dataset must either stream from the Chameleon Object Store (Swift), be staged from external object storage, or be re-downloaded at the start of each lease. Persistent images can be saved to either testbed's image repository, which preserves installed software but not bulk data.

Neither testbed permits 24/7 services. Reservations are finite, and a forgotten extension that lapses will tear the node down, return it to the pool, and wipe the local disk. Hardware is heterogeneous and refreshes year to year; hardware pages are worth checking monthly because retired node types simply disappear from the reservation calendar.

CloudLab's GitHub-hosted profiles are the dominant productivity multiplier. A well-written profile turns a multi-hour manual setup into a 90-second boot, and existing community profiles cover Kubernetes, Spark, OpenStack, RDMA testbeds, and many other common stacks. On Chameleon the equivalent artifact is a *Trovi notebook*, which bundles the lease, the image, and the analysis Jupyter pipeline into a single shareable record that another researcher can re-execute to reproduce a published figure.

GPU availability is the single biggest practical constraint on either platform. Network experiments and CPU-bound systems work scale almost arbitrarily; GPU experiments compete for a small inventory and benefit from booking off-peak (overnight, weekends) and from being prepared to fall back to an older accelerator generation when the latest tier is full.

Strategic Fit

CloudLab and Chameleon are outstanding fits for projects whose *research contribution is in the systems layer*. Distributed training-system benchmarks, scheduler designs, communication-library experiments, GPU-disaggregation studies, parallelism strategies, memory-hierarchy characterization, network-protocol prototyping, and reproducibility infrastructure all line up cleanly with the bare-metal, root-access, custom-image programming model. They are poor fits for projects whose only goal is to train a model end-to-end, since the SU economics, the lease ceilings, and the GPU inventory all penalize long, monolithic training runs.

The recommended pattern is to combine the testbeds with a production AI allocation. Train and validate the model on NAIRR or ACCESS where weeks-long jobs and large inventories are normal, then port the workload to Chameleon or CloudLab for instrumented profiling, ablations on parallelism strategy, communication-collective microbenchmarking, and any measurement that requires deterministic single-tenant hardware.

Summary

Platform	Hardware tier	Project-yr H200-hr eq.	Approval	Best fit
CloudLab Utah d740	A100 PCIe 40GB	1,400	1–2 wk	ML-systems
CloudLab Wisconsin c4130	4× V100 SXM2	2,200	1–2 wk	ML-systems / networking
CloudLab Clemson ibm8335	4× V100 + POWER9	2,200	1–2 wk	Systems / accelerator-arch
Chameleon CHI@UC a100_nvlink	4× A100 80GB	25,000	1–2 wk	ML-systems / training
Chameleon CHI@UC h100_nvlink	4× H100 SXM5	80,000	1–2 wk	ML-systems / training
Chameleon CHI@TACC Liqid H100	Up to 8× H100 PCIe (composable)	60,000–160,000	1–2 wk	GPU-disaggregation research

Platform	Hardware tier	Project-yr H200-hr eq.	Approval	Best fit
Chameleon CHI@UC gh200	1× GH200	20,000	1–2 wk	Grace-Hopper / unified-memory
Chameleon CHI@TACC rtx_6000	2× RTX 6000	5,300	1–2 wk	Inference / graphics

H200-hour equivalence is computed at dense FP16/BF16 tensor throughput (989 TFLOPS, 141 GB HBM3e, 4.8 TB/s) per H200 SXM GPU. Conversions below use the per-hour FP16 TFLOP ratio; memory-bound workloads scale differently and the bandwidth column is provided so readers can re-derive their own equivalence.

Sources

- CloudLab program and hardware: cloudlab.us · /hardware.php · docs.cloudlab.us
- CloudLab profiles and GitHub catalog: cloudlab.us/p · Emulab profile repos
- Chameleon Cloud program and hardware: chameleoncloud.org · /hardware · chameleoncloud.readthedocs.io
- Chameleon Trovi reproducibility catalog: Trovi share · Trovi docs
- Chameleon 2.0 AI hardware announcements: chameleoncloud.org/blog · news (H100 NVLink, GH200, MI210, Liqid H100 deployment posts, 2024–2026)
- NSF awards: NSF CNS-1743363 (CloudLab Phase II), NSF OAC-2229466 (Chameleon Phase 3 / "Chameleon 2.0", 2024–2028)
- GPU specs cross-checked against NVIDIA P100, V100, T4, RTX 6000, A100, H100, H200, GH200 datasheets and AMD MI100/MI210 product briefs.